

Higher Order Derivatives

Date_____ Period_____

For each problem, find the indicated derivative with respect to x .

1) $y = -x^2$ Find $\frac{d^2y}{dx^2}$

2) $f(x) = 4x^3$ Find f''

3) $y = -4x$ Find $\frac{d^3y}{dx^3}$

4) $f(x) = 5x^4$ Find f'''

5) $y = -5x^4$ Find $\frac{d^4y}{dx^4}$

6) $y = 3x^5 - 2x$ Find $\frac{d^3y}{dx^3}$

7) $y = -2x^3 - 4x^{-3}$ Find $\frac{d^3y}{dx^3}$

8) $y = -x^2 + 2\sqrt[5]{x^2}$ Find $\frac{d^3y}{dx^3}$

Critical thinking questions. Find the indicated derivatives with respect to x .

9) $y = 99x^{99}$ Find $\frac{d^{100}y}{dx^{100}}$

10) $f(x) = x^{99}$ Find $f^{(99)}$